Draft version June 29, 2025
Typeset using LaTeX **default** style in AASTeX7

A 9-Year TeV Gamma-Ray Survey of Active Galaxies by HAWC

The HAWC collaboration

ABSTRACT

Active galaxies constitute the majority of extragalactic TeV gamma-ray sources, often exhibiting large and rapid variability. However, the average TeV emission produced by these sources is poorly characterized. Thanks to its large duty cycle, the HAWC gamma-ray observatory can help address this issue. In this work, we present a renewed HAWC survey of active galaxies. A sample of 135 nearby ($z < 0.3$) gamma-ray emitting AGN, included in the Third Catalog of Hard Fermi-LAT Sources (3FHL) and located within 40° from HAWC's zenith, was analyzed using nine years of HAWC data (from 2014 to 2024). A power-law function with an EBL attenuation term was fitted to the HAWC spectra. Five sources presented solid detections (significance $> 5\sigma$), and 13 sources presented marginal detections (significance between 3σ and 5σ). These results demonstrate the capabilities of HAWC for detecting and characterizing the average VHE emission of AGN.

Keywords: Active galactic nuclei (16) — High Energy astrophysics (739) — Gamma-ray astronomy (628)

1. INTRODUCTION

Active galactic nuclei (AGN) are compact sources located in the center of some galaxies with a luminosity higher than that expected from stellar processes, which is driven by an accreting supermassive black hole (SMBH, $\sim 10^6 - 10^{11}$ $M_\odot$) (H. Netzer 2013). AGNs are the most common sources of extragalactic gamma rays, both at high energy (HE, $\sim 10^6 - 10^{11}$ eV) and very high energy (VHE, $> 10^{11}$ eV) bands. Among the multiple types of AGN, most of those that emit gamma rays are radio-loud AGNs, including BL Lac objects, flat-spectrum radio quasars (FSRQ), and radiogalaxies. Radio-loud AGNs usually display relativistic jets, which are considered to be the source of gamma-ray emission (R. Blandford et al. 2019).

AGNs are highly variable sources, with periods of high activity called flares (P. Padovani et al. 2017). These flares are particularly prominent at VHE, where blazars exhibit very fast variability down to 1 minute timescales (A. Pandey & C. Stalin 2022). Most of observations at these bands are carried out with imaging atmospheric Cherenkov telescopes (IACTs) (e.g. A. Abramowski et al. 2012; K. Abe et al. 2025). However, the relatively short duty cycle of these instruments has produced a bias in the studies of these sources toward flaring periods. Due to this bias, the average VHE emission of AGNs is poorly characterized, which complicates the understanding of the physical mechanisms that produce this complex emission. Thanks to their longer duty cycles, water Cherenkov instruments such as the High Altitude Water Cherenkov (HAWC) (A. Albert et al. 2020) and the Large High Altitude Air Shower Observatory (LHAASO) (Z. Cao et al. 2024), as well as the future Southern Wide-field Gamma-ray Observatory (SWGO) (J. Hinton 2021), can help to characterize the average VHE emission of these sources.

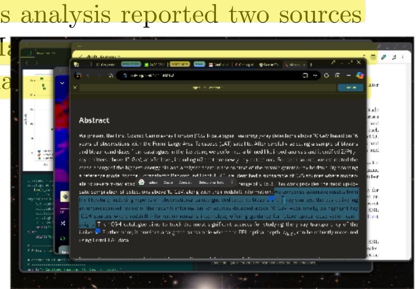
Problema

Posible solución

Observation of VHE emission from AGNs is also limited by the interaction of gamma rays with photons of the extragalactic background light (EBL), which is constituted by all the electromagnetic emission emitted by galaxies along the history of the Universe (A. Franceschini & G. Rodighiero 2017). This process causes an attenuation effect that increases with energy and redshift, limiting the detection of extragalactic VHE sources to $z \lesssim 0.3$.

In this work, we update a previously published analysis (A. Albert et al. 2021) that investigated a sample of HE gamma-ray-emitting AGNs included in the Third Catalog of Hard Fermi-LAT Sources (3FHL) (M. Ajello et al. 2017) using a 4.5-year VHE data set from the HAWC gamma-ray observatory. The previous analysis reported two sources with a solid detection (significance $> 5\sigma$), the BL Lac objects Markarian 421 and Ma… sources presented a marginal detection (significance between 3σ and 5σ), the radiogala…

No podemos detectar más allá de $z \lesssim 0.3$ por que se atenúa con el EBL.



objects 1ES 1215+303 and VER J0521+211. More detailed studies on some of these sources were presented in later works (A. Albert et al. 2022; R. Alfaro et al. 2022)

2. THE HAWC GAMMA-RAY OBSERVATORY

The HAWC gamma-ray and cosmic-ray observatory is located in Puebla, Mexico (latitude 18.995° N, longitude 97.308° W), near the Sierra Negra Mountain at an altitude of 4,100 meters above sea level. The instrument comprises an array of 300 Water Cherenkov Detectors (WCDs), each consisting of a tank that is 7.3 meters in diameter and 5 meters high, filled with purified water. Four photomultiplier tubes (PMTs) are positioned at the bottom of each WCD, which are triggered when secondary particles from air showers produce Cherenkov radiation within the WCD. Photon-hadron separation methods are employed to distinguish between air showers generated by cosmic rays and VHE gamma rays. In the case of gamma-ray events, various algorithms are utilized to estimate the direction and energy of the primary particle (A. Abeysekara et al. 2017a, 2019; A. Albert et al. 2024).

Estamos trabajando con un tipo de información característica NUEVA

El mejoramiento de el HAWC al tener ahora nos dá información a menor energía, por lo que

HAWC has been in continuous operation since November 2014, achieving a duty cycle of over 90%. It can observe 8.4 steradians every sidereal day, corresponding to approximately two-thirds of the sky. This instrument studies gamma rays across a wide energy range, from approximately 0.1 TeV to 100 TeV, enabling the investigation of a large number of phenomena (A. Abeysekara et al. 2017a, 2019).

In this work, we use the number of particle hits in the PMTs —that is, the size of the air shower— as a proxy for the primary particle energy. This method, referred to as FHIT, operates by dividing the events into 11 bins based on the fraction of PMTs activated during each event. Unlike in the previous survey, we now have access to the two lowest energy bins (B0 and B1), which are associated with the lowest energy photons that HAWC can detect. Due to EBL attenuation, the majority of the signal from VHE emitting AGNs is contained within these low-energy bins (A. Albert et al. 2024).

What's EBL?

3. EBL ATTENUATION OF GAMMA RAYS

All the electromagnetic radiation produced by galaxies throughout the history of the Universe constitutes the EBL (A. Franceschini & G. Rodighiero 2017). The spectral energy distribution (SED) of this radiation exhibits two peaks: one in the infrared and another in visible light. Extragalactic gamma rays interact with the EBL via pair production ($\gamma\gamma \rightarrow e^+e^-$). The survival probability of a gamma-ray photon is expressed as $e^{-\tau(E,z)}$, where $\tau(E,z)$ is the optical depth. The value of $\tau(E,z)$ increases with energy and redshift. This results in an attenuation effect that prevents extragalactic gamma rays from being detected at redshifts greater than $z \sim 0.3$ and limits the detected photon energies to lower values within the HAWC detection range (A. Albert et al. 2021) .

The condition for pair production to occur is $\omega > m_e c^2$, where m_e is the electron mass and c is the speed of light. The quantity ω is defined as

$$\omega = \sqrt{\frac{E_{ph} E_\gamma (1 - \cos\theta)}{2}}, \tag{1}$$

where E_γ is the gamma-ray energy, E_{ph} is the energy of the EBL photon, and θ is the angle of the $\gamma\gamma$ interaction. The maximum cross section for this interaction is achieved when $\omega \approx 1.4 m_e c^2$; therefore, gamma-ray photons with an energy of about 1 TeV interact with infrared EBL photons of approximately 0.5 eV. Moreover, interactions with the cosmic microwave background (CMB) occur primarily with the highest-energy gamma rays, rendering the study of extragalactic sources at energies around 100 TeV impossible.

The measurement and characterization of the EBL SED is a challenging task. Several models exist in the literature (e.g., A. Franceschini et al. 2008; J. D. Finke et al. 2010; A. Saldana-Lopez et al. 2021), each with various differences; however, these differences are not significant for the results reported in this paper. We utilized the model presented by A. Dominguez et al. (2011) to estimate the intrinsic gamma-ray spectra of the sources analyzed in this work.

4. SAMPLE, DATA AND METHODOLOGY

4.1. *Sample*

We use almost the same sample of nearby AGN as in the 2021 survey (A. Albert et al. 2021). This sample includes sources in the Third Catalog of Hard Fermi-LAT Sources (3FHL)(M. Ajello et al. 2017), which covers an energy range between 10 GeV and 2 TeV with observations from 2008 August 4 to 2015 August 2. This catalog has 1556 sources, of which 79% are identified or associated with extragalactic objects. We selected the survey sample using the following criteria:

- Sources in the 3FHL that are associated (positional coincident) or identified (confirmed by variability) with known Active Galactic Nuclei.

- Sources located at a redshift $z \leq 0.3$, a limit set by the EBL attenuation to detect sources in the TeV energy range. This criterion also excludes sources without redshift confirmation.

- Sources observed within 40° from the HAWC zenith.

It is necessary to look at the sources located near other HAWC sources to identify any possible contamination. To do that, we searched for objects in this sample located in the sky at $< 5°$ (about two times the angular resolution for the lowest HAWC bins) from sources reported in the Fourth HAWC Catalog (4HWC). Eight objects were found in this situation, which were:

- The Blazar Candidate of Uncertain type **PKS 0336-177 (3FHL J0339.2-1736)** and the BL Lac object **1ES 0347-121 (3FHL J0349.3-1159)** are located at 4.0° and 2.2° from the unassociated source 4HWC J0347-140.

- The BL Lac object **VER J0521+211 (3FHL J0521.7+2112)** lays at 3.1° from the Crab Nebula (4HWC J0534+2200).

- **RX J0648.7+1516 (3FHL J0648.7+1517)**, also a BL Lac object, lays close to several objects. Including 4HWC J0633+1719 (Geminga) at 4.1°, 4HWC J0701+1412 (2HWC J0700+143) at 3.3° and 4HWC J0634+1734 (Geminga Pulsar) at 4.2° from the BL lac object.

- The BL Lac objects **B3 1038+392 (3FHL J1041.7+3900)**, **RX J1100.3+4019 (3FHL J1100.3+4020)** and **5BZG J1105+3946 (3FHL J1105.8+3944)** are separated by 4.5°, 2.3° and 1.6 °, respectively, from Markarian 421 (4HWC J1104+3810).

- The BL Lac object **GB6 J1231+1421 (3FHL J1231.4+1422)** lays at 2.0° from the radio galaxy M87 (4HWC J1230+1223).

- The sky position of the BL Lac **SDSS J165249.92+402310.1 (3FHL J1652.7+4024)** is at only 0.7° from Markarian 501 (4HWC J1654+3944).

- The BL Lac object **1ES 2037+521 (3FHL J2039.4+5219)** is located in the sky at 4.5° from 4HWC J2108+5156 (LHAASO J2108+5157).

We did not discarded those sources in the previous list separated by more than 2.5° from bright HAWC sources, but we are taking them into account to interpret the final results. On the other hand, we excluded the sources 3FHL J1105.8+3944, 3FHL J1100.3+4020 and 3FHL J1652.7+4024 for being likely contaminated by Markarian 421 and Markarian 501. The final sample contains a total of 135 sources.

4.2. *Data and methodology*

We used the most recent HAWC fhit Pass 5 data set (A. Albert et al. 2024), which comprises nine years of HAWC data acquired from 2014 November 26 to 2024 January 17. The length of this time range doubles the 4.5 years of the previous survey. In addition to its larger time coverage, the Pass 5 analysis also presents the advantage of including two low-energy bins that were not part of the Pass 4 reconstructions (B0 and B1)(A. Albert et al. 2024). This is key for studying AGN, since their gamma-ray spectra are cut by EBL attenuation and most of the data correspond to low energies. However, low-energy bins also have low angular resolution (up to $\sim 2°$) (A. Albert et al. 2024).

For each of the 135 sources, we fit a function composed by a single power law and exponential term to the observed spectrum. The exponential term accounts for the EBL attenuation, and the power law represents the intrinsic spectral shape. The expression was the following:

$$\frac{\mathrm{d}N}{\mathrm{d}E} = K \left(\frac{E}{1\ \mathrm{TeV}} \right)^{-\alpha} e^{-\tau(E,z)}, \tag{2}$$

where K is the normalization, α is the spectral index, and $\tau(E, z)$ is the optical depth of the photon-photon gamma-ray attenuation by EBL. This work uses the EBL model of A. Dominguez et al. (2011). The spectral index was fixed to $\alpha = 2.5$ for the entire sample. This value corresponds to the spectral index used in the previous survey (A. Albert et al. 2021) as well as in 3HWC (A. Albert et al. 2020). This spectral index is also close to previously measured values for some relevant sources such as Mrk 421 ($\alpha = 2.26$ with a spectral cutoff at 5 TeV)(A. Albert et al. 2022), Mrk 501 ($\alpha = 2.61$) (A. Albert et al. 2022) and M87 ($\alpha = 2.63$) (R. Alfaro et al. 2022).

5. RESULTS

After the analysis, we compiled the TS and spectral normalization values (K) for each source. The results obtained were then used to test the null hypothesis, which states that the significance distribution tends to follow a Gaussian distribution (mean $\mu(s) = \langle s \rangle = 0$ and standard deviation $\sigma(s) = 1$), and that all differences between the two distributions are due to statistical fluctuations. This means that if the data were consistent with the null hypothesis, no evidence of gamma-ray emission would be found from this population. Additionally, we computed the 2σ upper bounds ($K_{2\sigma}$) for the normalization for each source.

Table 1 lists the best power law fits and significances for all sources in the sample. In addition, Table 2 shows the sources with marginal ($3\sigma - 5\sigma$) or confirmed ($> 5\sigma$) detections.

Significance

Distribución de la significancia

Table 1. HAWC Power-law Fits and Significances for the Sample of 135 AGN from the 3FHL Catalog

3FHL Source (1)	Counterpart (2)	Class (3)	Redshift (4)	TS (5)	$\pm\sqrt{TS}$ (6)	$K \pm \Delta K$ (7)	$K_{2\sigma}$ (8)
J0007.9+4711	RX J0007.9+4711	bll	0.28	9.4	-3.07	-18.4 ± 5.99	2.59
J0013.8-1855	RBS 0030	bll	0.0949	1.79	-1.34	-3.95 ± 2.96	2.69
J0018.6+2946	RBS 0042	bll	0.1	1.95	1.4	0.74 ± 0.53	1.81
J0037.8+1239	NVSS J003750+123818	bll	0.089	3.15	1.77	0.79 ± 0.44	1.68
J0047.9+3947	B3 0045+395	bll	0.252	0.52	-0.72	-2.01 ± 2.8	3.79
J0056.3-0936	TXS 0053-098	bll	0.1031	2.85	1.69	2.62 ± 1.55	5.71
J0059.3-0152	1RXS J005916.3-015030	bll	0.144	3.85	1.96	2.81 ± 1.43	5.67
J0112.1+2245	S2 0109+22	BLL	0.265	3.88	1.97	3.06 ± 1.55	6.18
J0123.0+3422	1ES 0120+340	bll	0.272	0.08	-0.29	-0.64 ± 2.23	3.84
J0131.1+5546	TXS 0128+554	bcu	0.0365	0.87	-0.93	-0.66 ± 0.71	0.83
J0152.6+0147	PMN J0152+0146	bll	0.08	2.14	1.46	0.8 ± 0.55	1.89
J0159.5+1047	RX J0159.5+1047	bll	0.195	0.99	0.99	1.2 ± 1.21	3.67
J0211.2+1051	MG1 J021114+1051	BLL	0.2	0.02	-0.16	-0.2 ± 1.25	2.32
J0214.5+5145	TXS 0210+515	bll	0.049	1.86	1.37	0.95 ± 0.7	2.37
J0216.4+2315	RBS 0298	bll	0.288	3.66	1.91	3.28 ± 1.72	6.7
J0217.1+0836	ZS 0214+083	bll	0.085	14.48	3.81	1.77 ± 0.47	2.7
J0219.1-1723	1RXS J021905.8-172503	bll	0.1287	0.69	0.83	3.76 ± 4.53	12.9
J0232.8+2017	1ES 0229+200	bll	0.14	1.92	-1.39	-1.0 ± 0.72	0.66
J0242.7-0002	NGC 1068	sbg	0.0038	1.21	-1.1	-0.1 ± 0.09	0.09
J0308.4+0408	NGC 1218	rdg	0.0288	6.2	2.49	0.49 ± 0.2	0.9
J0312.8+3614	V Zw 326	bll	0.071	1.07	1.04	0.45 ± 0.43	1.31
J0316.6+4120	IC 310	RDG	0.0189	0.56	0.75	0.13 ± 0.18	0.5
J0319.8+1845	RBS 0413	bll	0.19	6.42	2.53	2.67 ± 1.06	4.78
J0319.8+4130	NGC 1275	RDG	0.0176	0.29	0.54	0.09 ± 0.18	0.45

Table 1 *continued on next page*

Table 1 *(continued)*

3FHL Source (1)	Counterpart (2)	Class (3)	Redshift (4)	TS (5)	$\pm\sqrt{TS}$ (6)	$K \pm \Delta K$ (7)	$K_{2\sigma}$ (8)
J0325.6-1646	RBS 0421	bll	0.291	0.87	0.94	15.3 ± 16.4	48.3
J0326.3+0226	1H 0323+022	bll	0.147	3.73	1.93	2.25 ± 1.17	4.6
J0334.3+3920	4C+39.12	rdg	0.0203	0.45	0.67	0.11 ± 0.17	0.45
J0336.4-0348	1RXS J033623.3-034727	bll	0.1618	7.03	2.65	5.14 ± 1.94	9.0
J0339.2-1736	PKS 0336-177	bcu	0.0656	0.15	0.39	0.63 ± 1.63	3.87
J0349.3-1159	1ES 0347-121	bll	0.185	6.74	2.6	12.7 ± 4.9	22.5
J0416.8+0105	1ES 0414+009	bll	0.287	4.09	2.02	5.93 ± 2.94	11.8
J0424.7+0036	PKS 0422+00	bll	0.268	14.05	3.75	10.5 ± 2.8	16.0
J0521.7+2112	VER J0521+211	bll	0.108	21.78	4.67	2.47 ± 0.53	3.53
J0602.0+5316	GB6 J0601+5315	bcu	0.052	6.18	-2.49	-2.17 ± 0.87	0.46
J0617.6-1715	TXS 0615-172	bll	0.098	0.41	-0.64	-1.8 ± 2.81	3.95
J0648.7+1517	RX J0648.7+1516	bll	0.179	23.88	4.89	4.95 ± 1.01	7.01
J0650.7+2503	1ES 0647+250	bll	0.203	2.59	1.61	1.93 ± 1.2	4.31
J0656.2+4235	4C +42.22	bll	0.059	1.31	1.14	0.54 ± 0.48	1.51
J0725.8-0056	PKS 0723-008	bcu	0.127	0.47	0.69	0.79 ± 1.15	3.12
J0730.4+3307	1RXS J073026.0+330727	bll	0.112	0.58	0.76	0.53 ± 0.69	1.92
J0739.3+0137	PKS 0736+01	fsrq	0.191	0.89	-0.94	-1.62 ± 1.72	2.02
J0753.1+5354	4C +54.15	bll	0.2	2.33	-1.53	-11.0 ± 7.21	5.95
J0757.1+0957	PKS 0754+100	bll	0.266	0.24	0.49	0.88 ± 1.79	4.45
J0809.7+3457	B2 0806+35	bll	0.083	8.83	2.97	1.48 ± 0.5	2.49
J0809.8+5218	1ES 0806+524	bll	0.1371	0.65	0.81	2.67 ± 3.3	9.17
J0816.4+5739	SBS 0812+578	bll	0.054	2.77	-1.67	-2.02 ± 1.21	0.95
J0816.4-1311	PMN J0816-1311	bll	0.046	0.07	-0.26	-0.2 ± 0.77	1.36
J0816.9+2050	SDSS J081649.78+205106.4	bll	0.0583	3.91	1.98	0.53 ± 0.27	1.08
J0828.3+4153	GB6 B0824+4203	bll	0.2262	0.14	0.37	1.05 ± 2.87	6.81
J0831.8+0429	PKS 0829+046	bll	0.1738	1.58	1.26	1.66 ± 1.32	4.27
J0847.2+1134	RX J0847.1+1133	bll	0.1982	9.51	3.08	3.77 ± 1.22	6.23
J0850.6+3454	RX J0850.5+3455	bll	0.145	10.39	3.22	3.37 ± 1.05	5.45
J0908.9+2311	RX J0908.9+2311	bll	0.223	0.01	0.12	0.16 ± 1.3	2.74
J0912.4+1555	SDSS J091230.61+155528.0	bll	0.212	0.24	-0.49	-0.61 ± 1.23	1.88
J0930.4+4952	1ES 0927+500	bll	0.1867	0.07	0.26	1.15 ± 4.36	9.82
J1015.0+4926	1H 1013+498	bll	0.212	0.0	-0.02	-0.08 ± 5.07	10.1
J1027.0-1749	1RXS J102658.5-174905	bll	0.267	16.24	-4.03	-64.8 ± 16.1	5.4
J1041.7+3900	B3 1038+392	bll	0.2084	0.01	-0.08	-0.17 ± 2.1	4.06
J1053.6+4930	GB6 J1053+4930	bll	0.1404	0.03	-0.16	-0.43 ± 2.64	4.89
J1058.6+5628	TXS 1055+567	BLL	0.1433	0.02	-0.14	-0.72 ± 5.07	9.42
J1104.4+3812	Mrk 421	BLL	0.031	12584.78	112.18	31.3 ± 0.3	31.9
J1117.0+2014	RBS 0958	bll	0.138	0.04	0.19	0.14 ± 0.72	1.56
J1120.8+4212	RBS 0970	bll	0.124	1.31	1.15	1.42 ± 1.24	3.91

Table 1 *continued on next page*

Table 1 *(continued)*

3FHL Source (1)	Counterpart (2)	Class (3)	Redshift (4)	TS (5)	$\pm\sqrt{TS}$ (6)	$K \pm \Delta K$ (7)	$K_{2\sigma}$ (8)
J1125.9-0743	1RXS J112551.6-074219	bll	0.279	18.2	-4.27	-24.3 ± 5.71	1.84
J1136.8+2549	RX J1136.8+2551	bll	0.156	0.06	0.24	0.21 ± 0.88	1.97
J1140.5+1528	NVSS J114023+152808	bll	0.2443	2.38	1.54	2.24 ± 1.45	5.15
J1142.0+1546	MG1 J114208+1547	bll	0.299	0.15	0.38	0.69 ± 1.81	4.32
J1145.0+1935	3C 264	rdg	0.0216	7.87	2.81	0.37 ± 0.13	0.64
J1150.3+2418	OM 280	bll	0.2	0.47	0.69	0.8 ± 1.16	3.11
J1154.1-0010	1RXS J115404.9-001008	bll	0.2535	9.79	-3.13	-8.58 ± 2.74	1.17
J1204.2-0709	1RXS J120417.0-070959	bll	0.185	20.63	-4.54	-14.0 ± 3.07	0.94
J1217.9+3006	1ES 1215+303	bll	0.13	33.06	5.75	4.42 ± 0.77	5.95
J1219.7-0312	1RXS J121946.0-031419	bll	0.2988	11.79	-3.43	-14.4 ± 4.2	1.65
J1221.3+3010	PG 1218+304	bll	0.1837	43.19	6.57	7.84 ± 1.19	10.2
J1221.5+2813	W Comae	bll	0.1029	17.02	4.13	2.24 ± 0.55	3.34
J1224.4+2436	MS 1221.8+2452	bll	0.2187	18.04	4.25	5.5 ± 1.3	8.08
J1229.2+0201	3C 273	FSRQ	0.1583	9.32	-3.05	-4.01 ± 1.31	0.57
J1230.2+2517	ON 246	bll	0.135	5.35	2.31	1.68 ± 0.73	3.15
J1230.8+1223	M87	rdg	0.0042	38.11	6.17	0.46 ± 0.08	0.62
J1231.4+1422	GB6 J1231+1421	bll	0.2559	1.68	1.3	2.01 ± 1.55	5.12
J1231.7+2847	B2 1229+29	bll	0.236	5.19	2.28	3.52 ± 1.54	6.59
J1253.7+0328	MG1 J125348+0326	bll	0.0657	4.21	-2.05	-0.83 ± 0.4	0.25
J1256.2-1146	PMN J1256-1146	bcu	0.0579	6.48	-2.54	-2.22 ± 0.87	0.44
J1310.3-1158	TXS 1307-117	bll	0.14	17.51	-4.18	-12.9 ± 3.09	1.03
J1341.2+3959	RBS 1302	bll	0.1715	12.14	3.48	6.01 ± 1.73	9.41
J1402.6+1559	MC 1400+162	bll	0.244	1.38	1.17	1.69 ± 1.44	4.55
J1411.8+5249	SBS 1410+530	bcu	0.0765	0.24	0.48	0.67 ± 1.38	3.44
J1418.0+2543	1E 1415.6+2557	bll	0.2363	2.56	1.6	2.3 ± 1.44	5.2
J1419.4+0444	SDSS J141927.49+044513.7	bll	0.143	0.73	-0.85	-0.86 ± 1.01	1.27
J1419.7+5423	OQ 530	bll	0.1525	2.78	-1.67	-8.01 ± 4.8	3.67
J1428.5+4240	H 1426+428	bll	0.1292	8.97	2.99	4.06 ± 1.36	6.77
J1436.9+5639	RBS 1409	bll	0.15	4.11	-2.03	-11.3 ± 5.55	3.58
J1442.8+1200	1ES 1440+122	bll	0.1631	0.87	0.93	0.89 ± 0.95	2.8
J1449.5+2745	B2.2 1447+27	bll	0.2272	1.21	1.1	1.58 ± 1.44	4.47
J1500.9+2238	MS 1458.8+2249	bll	0.235	8.96	2.99	4.1 ± 1.37	6.85
J1508.7+2708	RBS 1467	bll	0.27	2.98	1.73	2.97 ± 1.72	6.4
J1512.2+0203	PKS 1509+022	fsrq	0.2195	0.15	-0.39	-0.78 ± 2.0	3.24
J1518.5+4044	GB6 J1518+4045	bll	0.0652	0.41	0.64	0.31 ± 0.49	1.3
J1531.9+3016	RX J1531.9+3016	bll	0.0653	1.95	1.4	0.46 ± 0.33	1.13
J1543.6+0452	CGCG 050-083	bcu	0.04	0.28	0.53	0.13 ± 0.24	0.61
J1554.2+2010	1ES 1552+203	bll	0.2223	13.88	3.73	4.73 ± 1.27	7.28
J1603.8+1103	MG1 J160340+1106	bll	0.143	0.0	0.01	0.01 ± 0.82	1.66

Table 1 *continued on next page*

Table 1 *(continued)*

3FHL Source (1)	Counterpart (2)	Class (3)	Redshift (4)	TS (5)	$\pm\sqrt{TS}$ (6)	$K \pm \Delta K$ (7)	$K_{2\sigma}$ (8)
J1615.4+4711	TXS 1614+473	fsrq	0.1987	0.33	-0.57	-2.11 ± 3.69	5.48
J1643.5-0646	NVSS J164328-064619	bcu	0.082	0.71	-0.84	-0.77 ± 0.91	1.15
J1647.6+4950	SBS 1646+499	bll	0.0475	1.4	1.18	0.71 ± 0.6	1.94
J1653.8+3945	Mrk 501	BLL	0.033	843.66	29.05	7.6 ± 0.29	8.18
J1719.2+1745	PKS 1717+17	bll	0.137	3.55	1.88	1.34 ± 0.71	2.76
J1725.4+5851	7C 1724+5854	bll	0.297	13.55	-3.68	-71.7 ± 19.5	7.14
J1728.3+5013	I Zw 187	bll	0.055	0.0	0.04	0.03 ± 0.72	1.5
J1730.8+3715	GB6 J1730+3714	bll	0.204	0.0	-0.02	-0.04 ± 1.83	3.64
J1744.0+1935	S3 1741+19	bll	0.083	15.39	3.92	1.49 ± 0.38	2.27
J1745.6+3950	B2 1743+39C	bll	0.267	0.32	0.57	1.73 ± 3.04	7.91
J1813.5+3144	B2 1811+31	bll	0.117	0.01	-0.07	-0.05 ± 0.69	1.34
J1917.7-1921	1H 1914-194	bll	0.137	4.57	-2.14	-12.0 ± 5.6	3.47
J2000.4-1327	NVSS J200042-132532	fsrq	0.222	0.01	0.12	0.9 ± 7.5	16.0
J2014.4-0047	PMN J2014-0047	bll	0.231	1.12	1.06	2.68 ± 2.53	7.72
J2039.4+5219	1ES 2037+521	bll	0.054	0.81	-0.9	-0.74 ± 0.82	0.98
J2042.0+2428	MG2 J204208+2426	bll	0.104	15.07	3.88	1.97 ± 0.51	3.0
J2055.0+0014	RGB J2054+002	bll	0.1508	0.82	0.9	1.21 ± 1.34	3.91
J2108.8-0251	TXS 2106-030	bll	0.149	0.99	0.99	1.59 ± 1.6	4.82
J2143.5+1742	OX 169	fsrq	0.211	7.67	2.77	3.3 ± 1.19	5.7
J2145.8+0718	MS 2143.4+0704	bll	0.235	0.04	0.2	0.34 ± 1.69	3.66
J2150.2-1412	TXS 2147-144	bll	0.229	0.04	0.19	1.64 ± 8.53	18.6
J2202.7+4216	BL Lacertae	BLL	0.069	0.15	0.39	0.22 ± 0.56	1.36
J2250.0+3825	B3 2247+381	bll	0.119	0.01	-0.07	-0.07 ± 0.92	1.79
J2252.0+4031	MITG J2252+4030	bll	0.229	3.46	-1.86	-4.83 ± 2.6	1.72
J2314.0+1445	RGB J2313+147	bll	0.1625	2.05	1.43	1.28 ± 0.9	3.09
J2322.6+3436	TXS 2320+343	bll	0.098	2.3	1.52	0.91 ± 0.6	2.12
J2323.8+4210	1ES 2321+419	bll	0.059	3.62	-1.9	-0.88 ± 0.46	0.31
J2329.2+3755	NVSS J232914+375414	bll	0.264	0.01	-0.07	-0.2 ± 2.61	4.99
J2338.9+2123	RX J2338.8+2124	bll	0.291	1.29	1.14	1.94 ± 1.71	5.34
J2346.6+0705	TXS 2344+068	bll	0.172	10.3	3.21	3.74 ± 1.17	6.06
J2347.0+5142	1ES 2344+514	bll	0.044	2.24	1.5	0.95 ± 0.64	2.26
J2356.2+4035	NVSS J235612+403648	bll	0.131	0.71	-0.84	-1.01 ± 1.2	1.47
J2359.3-2049	TXS 2356-210	bll	0.096	0.01	-0.11	-0.36 ± 3.29	6.26

Note—In column (3), associations are indicated in lowercase and identifications in uppercase. Column (4) lists the redshifts reported in 3FHL, which were used in this analysis. TS and $\pm\sqrt{TS}$, listed in columns (5) and (6), were obtained allowing fluxes to be positive or negative. The sign in $\sqrt{TS}$ corresponds to the normalization K. The units of K in columns (7) are $\mathrm{TeV}^{-1}\mathrm{cm}^{-2}\mathrm{s}^{-1}$. $K_{2\sigma}$ are the 2 σ upper bounds obtained following G. J. Feldman & R. D. Cousins (1998).

Figure 1 depicts a histogram showing the significance distribution (defined as $s = \pm\sqrt{TS}$) for the 135 sources in the clear sample. It can be noticed that two sources have much higher significance than the rest, Mrk 421 and Mrk

Table 2. Marginally detected, with significance between 3σ and 5σ, and confirmed detected, with significance $> 5\sigma$, sources in the survey of active galaxies.

Detections ($> 5\sigma$)	TS ($\sqrt{TS}$)	z	Marginal detections ($3\sigma - 5\sigma$)	TS ($\sqrt{TS}$)	z
Mrk 421	12584.78 (112.18)	0.031	**ZS 0214+083**	14.48 (3.81)	0.085
1ES 1215+303	33.06 (5.75)	0.13	**PKS 0422+00**	14.05 (3.75)	0.268
PG 1218+304	43.19 (6.57)	0.1837	VER J0521+211	21.78 (4.67)	0.108
M87	38.11 (6.17)	0.0042	RX J0648.7+1516	23.88 (4.89)	0.179
Mrk 501	843.66 (29.05)	0.033	RX J0847.1+1133	9.51 (3.08)	0.1982
			RX J0850.5+3455	10.39 (3.22)	0.145
			W Comae	17.02 (4.13)	0.1029
			MS 1221.8+2452	18.04 (4.25)	0.2187
			RBS 1302	12.14 (3.48)	0.1715
			1ES 1552+203	13.88 (3.73)	0.2223
			S3 1741+19	15.39 (3.92)	0.083
			MG2 J204208+2426	15.07 (3.88)	0.104
			TXS 2344+068	10.3 (3.21)	0.172

NOTE—Those source that are not reported in TeVCat are written in bold.

501. The mean of the sample is $\mu = 1.75\sigma$ with a standard deviation of $\sigma = 10.03$. As mentioned, under the null hypothesis, the distribution of significances tends to a Gaussian distribution, which would imply that these data do not contain evidence of gamma-ray emission from this population. The p-value is the probability of obtaining a test value (random value from the distribution) that is at least as extreme as the observed value. This quantity is used to determine whether the null hypothesis is rejected ($p_v < 10^{-3}$) or whether it is consistent with the data. In this case, the p-value was $p_v = 4 \times 10^{-92}$, which definitely rejects the null hypothesis.

To better understand this population, Figure 2 shows a significant histogram excluding Mrk 421 and Mrk 501. The shape of the histogram resembles a normal distribution, but slightly shifted to the right. In this case, the mean is $\mu = 0.72$ and the standard deviation is $\sigma = 2.12$. Finally, the p-value increases with respect to the previous value to $p_v = 5 \times 10^{-17}$, which is expected by excluding the two most significant sources. However, the null hypothesis is still excluded from this p-value, which is strong evidence that we have detected Tev gamma-ray emission from this population.

Finally, Table 3 presents the flux normalization along with statistical and systematic uncertainties for all sources exceeding the 3σ threshold. We calculated the systematic uncertainties following the procedures outlined in previous works (A. Albert et al. 2024; A. Abeysekara et al. 2019, 2017a).

6. DISCUSSION

After analyzing a sample of 135 nearby Active Galactic Nuclei, we obtained a p-value of $p_v = 4 \times 10^{-92}$ for the TS distribution. Since $p_v < 10^{-3}$ (W. H. Press et al. 2007), we reject the null hypothesis, which states that all deviations from the normal distribution in the significances are due to statistical fluctuations rather than actual emissions from these sources. When excluding the most significant sources (Mrk 421 and Mrk 501), we obtained a p-value of $p_v = 5 \times 10^{-17}$, further rejecting the null hypothesis in this case.

We found 18 sources with significance $s > 3\sigma$, representing a substantial increase compared to the five sources identified in the 4.5-year survey. Of these 18 sources, five exhibited confirmed detections with $s > 5\sigma$, specifically four BL Lac objects (Mrk 421, Mrk 501, PG 1218+304, and 1ES 1215+303) and one radio galaxy (M87). The remaining 13 sources, all of which are BL Lacs, are considered marginal detections with significance $3\sigma < s < 5\sigma$. Seven of these marginally detected sources are not listed in TeVCat, suggesting that they may be newly discovered TeV sources. Two marginal detections from the previous survey were confirmed as solid detections in this analysis: the radio galaxy

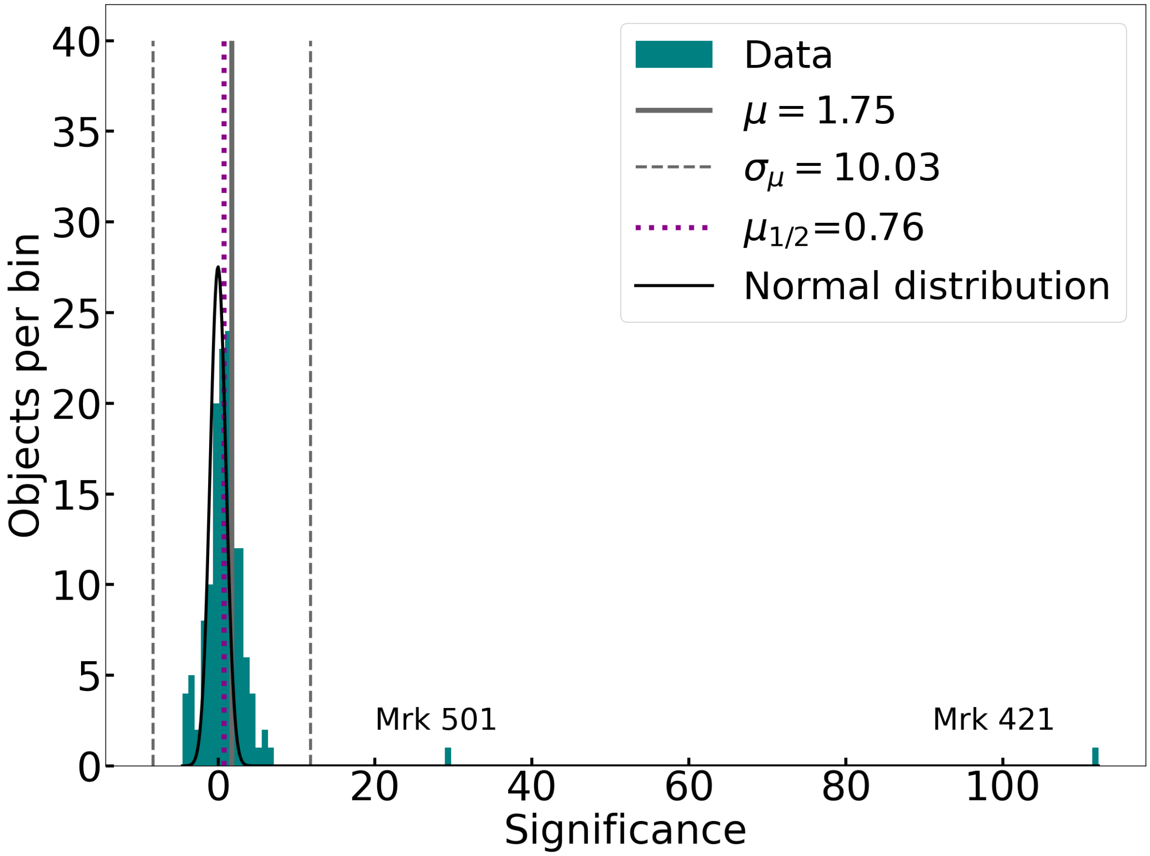


Figure 1. Histogram of significance (s=$\sqrt{TS}$) for the sample of 135 nearby AGN. The black curve depicts a normal distribution centered in $s = 0$ with the standard deviation zero for the the given number of objects. The solid gray vertical line shows the position of the mean ($\mu(s)$) while the dashed gray lines represent $\mu(s) \pm \sigma(s)$, where $\sigma(s)$ is the standard deviation. Finally, the dotted line depicts the sample's median. The significance values of Mkr 421 and Mrk 501 are labeled in the plot.

M87 and the BL Lac object 1ES 1215+303. The other new source with significance $s > 5\sigma$, the BL Lac object PG 1218+304, was below the 3σ threshold in the previous survey. However, it is located only 0.88° from another TeV BL Lac, 1ES 1215+303. Therefore, a more detailed analysis is necessary to investigate potential cross-contamination between the two sources.

A more detailed view of each of the 18 detected sources is provided below:

6.1. *Confirmed detections*

- **Markarian 421 (Mrk 421)** This source (Coordinates: R.A. 11h 04m 19s, Dec. +38° 11' 41") is the brightest extragalactic object at TeV energies. It is located at a redshift of $z = 0.031$ and has been extensively studied in gamma-ray bands. Both its TeV flux and spectrum have been reported to be highly variable (J. Albert et al. 2007; M. L. Ahnen et al. 2016). The intrinsic spectrum (corrected for EBL absorption) of this source shows evidence of an exponential cut-off (J. Albert et al. 2007), which was confirmed by long-term HAWC observations (A. Albert et al. 2022).

  In the previous survey, a test statistic of $TS = 4167$ (64.55 σ) was reported. This work obtains a $TS = 12584.78$ (112.18 σ). This represents a significant increase ($\Delta TS = 8417$) compared to the previous work. This increase is expected due to the additional exposure time in this survey and the improvements in the HAWC analysis implemented in Pass 5, primarily due to enhanced HAWC sensitivity at the lowest energies.

  The spectral normalization obtained in this work for this source is $K = (31.3 \pm 0.3) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1}, which is consistent within 1σ with the value reported in the previous survey, $K = (29.5 \pm 0.5) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1}. Due to the high variability of this object, this agreement indicates that the HAWC data are constraining the average VHE emission of this source.

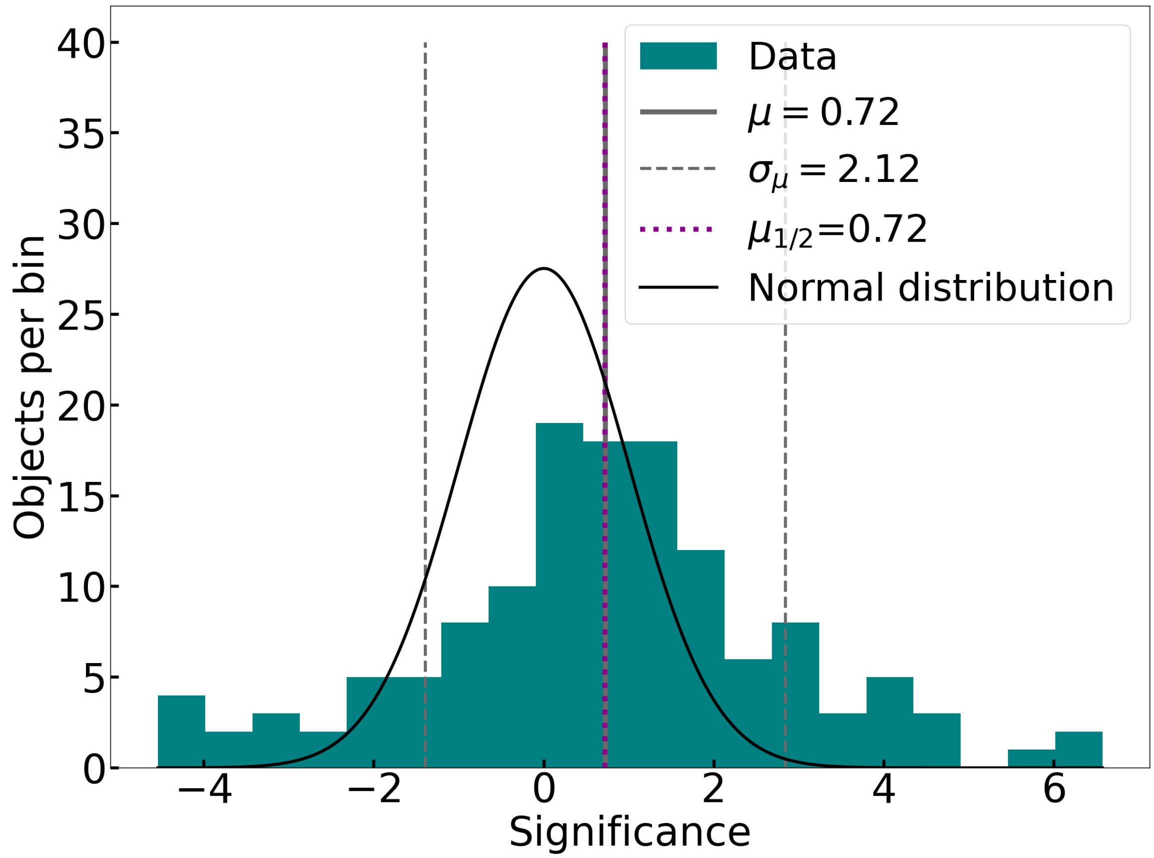


Figure 2. Histogram of significance (s=$\sqrt{TS}$) excluding Mrk 421 and Mrk 501. The black curve depicts a normal distribution centered in $s = 0$ with the standard deviation zero for the the given number of objects. The solid gray vertical line shows the position of the mean ($\mu(s)$) while the dashed gray lines represent $\mu(s) \pm \sigma(s)$, where $\sigma(s)$ is the standard deviation. Finally, the dotted line depicts the median, which is located very close to the mean of this subsample.

- **Markarian 501 (Mrk 501)**

This BL Lac object (Coordinates: R.A. 16h 53m 52.2s, Dec. +39° 45' 37", redshift $z = 0.033$) is a well-known and extensively studied gamma-ray source. Like many other TeV blazars, it exhibits variable flux and variable spectral index (V. Acciari et al. 2011). A detailed long-term analysis by HAWC was published for this source and Mrk 421 (A. Albert et al. 2022).

The test statistic obtained in this work, $TS = 843.66$ (29.05 σ), represents an increase of $\Delta TS = 566.7$ compared to the 4.5-year survey ($TS = 276.97$ (16.64 σ)). This result reverses a trend reported in previous HAWC studies, which indicated that the significance of this source was decreasing over time due to its high variability (A. Albert et al. 2021; A. Abeysekara et al. 2017b).

The normalization of the VHE spectrum obtained in this work for Mrk 501 is $K = (7.60 \pm 0.29) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1}, which is consistent within 1 σ with the value reported in the 4.5-year survey, $K = (7.74 \pm 0.49) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1}. As mentioned above, this consistency provides evidence of HAWC's capacity to constrain the average VHE emission from these sources.

- **M87**

This object (Coordinates: R.A. 12h 30m 47.2s, Dec. +12° 23' 51", redshift $z = 0.0044$) is classified as an FR-I radio galaxy. It is located at a distance of 16.4 Mpc, and its SMBH was imaged with the Event Horizon Telescope (EHT) (T. E. H. T. Collaboration et al. 2019). It was the first of its kind to be detected at VHE (F. Aharonian et al. 2003) and has exhibited several TeV flares since it was first observed (A. Abramowski et al. 2012). A spectral hardening of its gamma-ray spectrum is reported at approximately 10 GeV (F. A. Benkhali

Table 3. Flux normalization, statistical errors and systematic uncertainties for all sources exceeding the 3 σ threshold

Source	Flux normalization (K)	Statistical uncertainty (ΔK)	Systematic uncertainty
ZS 0214+083	1.77	± 0.47	$^{+0.07}_{-0.19}$
PKS 0422+00	10.5	± 2.79	$^{+0.7}_{-1.41}$
TXS 0518+211	2.47	± 0.53	$^{+0.13}_{-0.29}$
RX J0648.7+1516	4.95	± 1.01	$^{+0.18}_{-0.38}$
RX J0847.1+1133	3.77	± 1.22	$^{+0.16}_{-0.24}$
RX J0850.5+3455	3.37	± 1.05	$^{+0.07}_{-0.23}$
RX J1100.3+4019	11.6	± 2.54	$^{+0.28}_{-0.65}$
Mrk 421	31.3	± 0.33	$^{+1.54}_{-4.02}$
1ES 1215+303	4.42	± 0.77	$^{+0.12}_{-0.56}$
PG 1218+304	7.84	± 1.19	$^{+0.23}_{-0.67}$
W Comae	2.24	± 0.55	$^{+0.13}_{-0.28}$
MS 1221.8+2452	5.5	± 1.3	$^{+0.2}_{-0.33}$
M87	0.46	± 0.08	$^{+0.08}_{-0.07}$
RBS 1302	6.01	± 1.73	$^{+0.14}_{-0.5}$
1ES 1552+203	4.73	± 1.27	$^{+0.16}_{-0.29}$
Mrk 501	7.6	± 0.29	$^{+0.53}_{-1.07}$
S3 1741+19	1.49	± 0.38	$^{+0.06}_{-0.22}$
MG2 J204208+2426	1.97	± 0.51	$^{+0.04}_{-0.2}$
TXS 2344+068	3.74	± 1.17	$^{+0.08}_{-0.38}$

Note—Flux units 10^{-12} TeV^{-1}cm^{-2}s^{-1}.

et al. 2019), which could be associated with an emission component (in addition to the Synchrotron Self-Compton scenario) dominating the VHE emission. In R. Alfaro et al. (2022), a photohadronic model is fitted to its HAWC spectrum to explain this phenomenon. This source presented a $TS = 38.11$ (6.17 σ), representing an increase of $\Delta TS = 25.18$ compared to the previous work, confirming its detection by HAWC. The spectral normalization obtained in this work is $K = (0.46 \pm 0.08) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1}, which is consistent within 1 σ with the value reported in the previous survey, $K = (0.56 \pm 0.16) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1}.

- **1ES 1215+303**

The blazar 1ES 1215+303 (Coordinates: R.A. 12h 17m 48.5s, Dec. +30° 06' 06") is classified as a BL Lac object. The redshift of this blazar has been confirmed to be $z \sim 0.13$ by various teams using different methods. For example, S. Paiano et al. (2017) reported detections of the emission lines [O II] 3727 Å and [O III] 5007 Å, observed with the 10-meter Gran Telescopio de Canarias (GTC). These measurements allowed them to report a redshift of $z = 0.131$. Additionally, a redshift measurement of $z = 0.1305 \pm 0.0030$ was reported in A. Furniss et al. (2019), based on the detection of the Ly α line at 1374 Å using the HST Cosmic Origin Spectrograph (COS).

1ES 1215+303 is a well-known and extensively studied gamma-ray source, which was discovered at VHE by MAGIC (J. Aleksić et al. 2012), and VERITAS conducted its most detailed VHE monitoring (J. Valverde et al. 2020). This source displayed its most prominent VHE flare in 2014, which was observed by VERITAS (A. Abeysekara et al. 2017c) and reached a flux of 2.4 Crab units. The maximum energy of these observations is not always reported by IACT collaborations (e.g., J. Valverde et al. 2020), likely due to its large uncertainty.

This source was reported as a marginal detection in the previous survey with $TS = 11.36$ (3.37 σ). In this work, a $TS = 33.06$ (5.75 σ) is obtained, confirming its detection. The flux normalization obtained for this source is

$K = (4.42 \pm 0.77) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1}, which is consistent with the value obtained in the previous survey, $K = (4.64 \pm 1.38) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1}. However, since the blazar PG 1218+304 is only 0.88° from this source, it is likely that there is cross-contamination between these two sources.

- **PG 1218+304**

The blazar PG 1218+304 (Coordinates: R.A. 12h 21m 26.3s, Dec. +30° 11' 29") is classified as a BL Lac object. The redshift of this object ($z = 0.182$) was determined by N. Bade et al. (1998), using spectroscopic measurements of absorption features in the host galaxy. A similar value, $z = 0.1836$, is reported by the Sloan Digital Sky Survey (SDSS) [1]. The spectral energy distribution (SED) of this object also shows a double-peaked structure, and one-zone SSC scenarios have been used to model its emission from the optical to the gamma-ray bands (N. Sahakyan 2020; K. Singh et al. 2019).

PG 1218+304 was discovered as a TeV gamma-ray emitter by MAGIC (J. Albert et al. 2006), whose observations were taken during a non-flaring state and reported a non-variable gamma-ray flux of approximately 2×10^{-11} erg cm^{-2} s^{-1} at 100 GeV. Subsequent IACT results described this object as a variable source in those bands, ranging from approximately 6% to approximately 20% of the Crab flux (V. Acciari et al. 2010; S. O. Brien 2019).

This work reports a test statistic of $TS = 43.19$ ($6.57\ \sigma$) for this source. This result represents a significant change compared to $TS = 5.02$, reported in the previous survey. The spectral normalization obtained in this analysis, $K = (7.84 \pm 1.19) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1}, is consistent within 1 σ with the value obtained in the previous survey, $K = (5.23 \pm 2.34) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1}, in which the source was not even marginally detected. Finally, as mentioned, since this source is located 0.88° from 1ES 1215+303, a more detailed analysis is needed to investigate the likely cross-contamination between these two sources.

Marginal detections

- **ZS 0214+083**

This object (Coordinates: R.A. 02h 17m 17.1249s, Dec. +08° 37' 03.898") is classified as a BL Lac object. In M. S. Shaw et al. (2013), a redshift of $z = 0.085$ is reported for this object, which was obtained by detecting features of the host galaxy. A previous redshift measurement of $z = 1.4$ was reported in A. Gorshkov & V. Konnikova (1993), based on a possible broad emission observed in the spectrum. However, the authors warned that these features could be misidentified due to the low signal-to-noise ratio in the spectrum. This source is not reported by TeVCat, and the closest object in that catalog is the BL Lac object RGB J0152+017, located 9.2° away. In this work, a $TS = 14.48$ ($3.81\ \sigma$) is obtained, corresponding to an increase of $\Delta TS = 9.90$ compared to the previous survey. The normalization reported in this work is $K = (1.77 \pm 0.47) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1}, which is consistent within 1 σ with the value reported in the 4.5-year survey, $K = (1.70 \pm 0.80) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1}.

- **VER J0521+211**

This source (Coordinates: R.A. 05h 21m 45s, Dec: +21° 12' 51.4") is a VHE emitter classified as a BL Lac object. Several different estimates have been provided for its redshift. In M. S. Shaw et al. (2013), a redshift of $z = 0.108$ is reported based on a weak emission line at 5940 Å attributed to [N II] 6583 Å. This value is used in this work; however, it could not be confirmed by S. Archambault et al. (2013) and S. Paiano et al. (2017), who provided a lower limit of $z > 0.18$. Using its TeV emission, C. Adams et al. (2022) estimated an upper limit of $z < 0.28$.

This object is a well-known TeV source (S. Archambault et al. 2013; C. Adams et al. 2022), for which a $TS = 9.49$ ($3.08\ \sigma$) was reported in the original survey. In this work, a $TS = 21.78$ ($4.67\ \sigma$) is obtained, corresponding to an increase of $\Delta TS = 12.29$ compared to the previous value. Its spectral normalization reported in this work is $K = (2.47 \pm 0.53) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1}, which is consistent within 1σ with the value reported in the previous survey, $K = (2.85 \pm 0.93) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1}.

[1] http://www.sdss.org/dr13/data_access/bulk/

As mentioned, this source is located in the sky only 3.1° from one of the brightest TeV sources, the Crab Nebula, which is likely contaminating this blazar's emission. Due to this proximity and the uncertainty in its redshift, a more detailed analysis is necessary to precisely characterize the long-term emission observed by HAWC.

- **RX J0648.7+1516** → Una de las más significativas TS=23.8

This blazar (Coordinates: R.A. 06h 48m 45.6s, Dec: +15° 16' 12") is classified as a BL Lac object. In E. Aliu et al. (2011), a redshift of $z = 0.179$ is determined for this object, based on several absorption lines detected in the host galaxy spectrum. VERITAS reported the first TeV detection of this object (R. A. Ong et al. 2010). In this work, a $TS = 23.88$ (4.89 σ) is obtained, making it one of the most significant sources in this sample. Compared to the previous survey, this source reports a $\Delta TS = 23.62$, indicating a significant change. However, it is worth mentioning that this object is located near some bright HAWC sources, including the Geminga region. A more detailed analysis is necessary to remove any possible contamination from those other TeV emitters.

- **RX J0847.1+1133**

This BL Lac object (Coordinates: R.A. 08h 47m 12.9s, Dec: +11° 33' 50") is classified as a known TeV source. It has a redshift of $z = 0.198$ as reported by the Sloan Digital Sky Survey (C. P. Ahn et al. 2012). This source was discovered in the TeV bands by MAGIC (R. Mirzoyan 2014). We report a $TS = 9.51$ (3.08 σ). This indicates that the source has an increase in TS of $\Delta TS = 9.42$ compared to the previous survey. The value of the spectral normalization reported in this work is $K = (3.77 \pm 1.22) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1}. Despite its large uncertainty, it is consistent within 1 σ with the value from the previous survey, $K = (0.72 \pm 2.45) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1}.

- **RX J0850.5+3455**

This blazar (Coordinates: R.A. 08h 50m 36.1958s, Dec: +34° 55' 22.716") is classified as a BL Lac object. It has a redshift of $z = 0.145$ as measured by the Sloan Digital Sky Survey (C. P. Ahn et al. 2012). This source has no previous reports of detection in the TeV bands. We obtained a $TS = 10.39$ (3.22 σ), which corresponds to a $\Delta TS = 8.96$ compared to the previous survey. The spectral normalization reported in this analysis is $K = (3.37 \pm 1.05) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1}. This value is consistent with the value obtained in the previous survey, $K = (2.33 \pm 1.95) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1}.

- **W Comae**

This BL Lac object (Coordinates: R.A. 12h 21m 31.7s, Dec: +21° 13' 59") is a known TeV emitter, first discovered by VERITAS (V. Acciari et al. 2008). The redshift of this source was determined to be $z = 0.102$ in S. Paiano et al. (2017) by detecting the emission lines corresponding to [O III] 5007 Å and Hα at that redshift. Moreover, the absorption features of the host galaxy, including the Ca II (3934, 3968 Å) doublet, the G-band at 4305 Å, and Mg I at 5175 Å, were also detected at $z = 0.102$. This result confirms a previous measurement by D. Weistrop et al. (1985).

We report a $TS = 17.02$ (4.13 σ) for this source, which implies a $\Delta TS = 10.99$ from the previous survey. The normalization values obtained in both surveys ($K = (2.24 \pm 0.55) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1} for this one, and $K = (2.33 \pm 0.95) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1} for the previous one) agree within 1 σ.

This source, along with other objects such as 1ES 1215+303, PG 1218+204, and MS 1221.8+2452, is part of the Coma Cluster region. Due to the presence of many TeV sources in this area, a dedicated analysis of this sky zone is warranted.

- **MS 1221.8+2452**

This BL Lac object (Coordinates: R.A. 12h 24m 24.2s, Dec: +24° 36' 24") is a well-known TeV source located in the Coma cluster region. S. Paiano et al. (2017) estimated a redshift of $z = 0.218$ using the Ca II doublet and G-band 4305 Å absorption lines, as well as possible emission lines of Hα and [N II] 6583 Å. This result confirms previous tentative estimates by S. L. Morris et al. (1991) and T. A. Rector et al. (2000). The test statistic reported for this source in this work is $TS = 18.04$ (4.25 σ), representing a $\Delta TS = 17.49$ compared to the last survey. The normalization values obtained in both surveys ($K = (5.5 \pm 1.3) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1} for this one, and $K = (1.99 \pm 2.68) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1} for the previous one) agree within 1σ.

• RBS 1302

This source (Coordinates: R.A. 13h 41m 04.92s, Dec: +39° 59' 35.16") is classified as a BL Lac object. Its redshift is $z = 0.1715$, as reported by the Sloan Digital Sky Survey (SDSS). This object had not previously been reported as a TeV emitter, and its nearest TeVCat source is the BL Lac object H 1426+428, located at 9.3° away. A $TS = 12.14$ (3.48 σ) is obtained for this source, which corresponds to a $\Delta TS = 13.11$ compared to the last survey, in which a negative significance was reported ($TS = -0.97$ (-0.99 σ)). The value of the spectral normalization obtained in this work is $K = (6.01 \pm 1.73) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1}, which, as expected, is not consistent with the negative value reported in the previous survey, $K = (-3.35 \pm 3.40) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1}.

• 1ES 1552+203

The coordinates of this BL Lac object are R.A. 15h 54m 24.1302s, Dec: +20° 11' 25.11". It has a redshift of $z = 0.222$, as measured by the Sloan Digital Sky Survey. It is not reported as a TeV emitter, and its closest source reported in TeVCat is the BL Lac object PG 1553+113, located 9° away. A $TS = 13.88$ (3.73 σ) is measured for this source, corresponding to an increase in TS of $\Delta TS = 12.12$ compared to the 4.5-year survey. The value of the spectral normalization is $K = (4.73 \pm 1.27) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1}, which is not consistent within 1σ with the value reported in the previous survey, $K = (1.76 \pm 1.33) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1}.

• S3 1741+19

The coordinates of this BL Lac object are R.A. 17h 44m 01.2s, Dec: +19° 32' 47". Its redshift is $z = 0.084$, as reported by J. Heidt et al. (1999). This value was determined by measuring the locations of the Ca K+H, G-band, Mg b, and Na D absorption lines. This source was first discovered at very high energy by MAGIC (M. L. Ahnen et al. 2017).

A $TS = 15.39$ (3.92 σ) is reported for this source in this work, corresponding to a $\Delta TS = 13.77$ compared to the previous survey, where $TS = 1.62$ (1.27 σ). The spectral normalization obtained in this work is $K = (1.49 \pm 0.38) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1}, which is consistent within 1σ with the value reported in the 4.5-year survey, $K = (0.83 \pm 0.66) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1}.

As mentioned, this source is located only 4.6° from 3HWC J1743+149, an unidentified class source. More detailed analysis is needed to determine whether these two sources are cross-contaminated.

• MG2 J204208+2426

The coordinates of this BL Lac object are R.A. 20h 42m 06.0500s, Dec: +24° 26' 52.340". The redshift of this source is reported as $z = 0.104$ by M. S. Shaw et al. (2013) and was measured using absorption features of the host galaxy. This object has not previously been reported as a TeV emitter, and its nearest TeVCat source is the unidentified class source 3HWC J2023+324, located 8.99° away. In this work, a $TS = 15.07$ (3.88 σ) is reported for this object, representing an increase of $\Delta TS = 14.49$. The value of the spectral normalization is $K = (1.97 \pm 0.51) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1}, consistent within 1σ with the value reported in the previous survey, $K = (0.67 \pm 0.89) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1}.

• TXS 2344+068

The coordinates of this BL Lac object are R.A.: 23h 46m 39.9333s, Dec: +07° 05' 06.846". Its redshift has been reported to be $z = 0.172$, as measured from SDSS observations. This object is not reported as a TeV emitter, and its nearest TeVCat source is the BL Lac object RGB J2243+203, located at 20.2°. In this work, a $TS = 10.3$ (3.21σ) is reported for this object. This value represents a $\Delta TS = 8.06$ compared to $TS = 2.24$ (1.5σ)reported in the previous survey. The spectral normalization obtained in this work is $K = (3.74 \pm 1.17) \times 10^{-12}$ TeV^{-1} cm^{-2} s^{-1}, which is consistent within 1σ with the value reported in the 4.5-year survey.

7. SUMMARY AND CONCLUSIONS

A sample of 135 nearby ($z < 0.3$) Active Galactic Nuclei from the Third Catalog of Hard Fermi-LAT Sources (3FHL) (M. Ajello et al. 2017) was analyzed using nine years of HAWC data, covering the period from November 2014 to January 2024. A single power law with an exponential term to account for gamma-ray attenuation by photon-photon interactions with the extragalactic background light (EBL) was fitted to the entire sample. During this process, the spectral index was set to $\alpha = 2.5$, and the normalization was treated as a free parameter. According to the null hypothesis, the differences between a Gaussian distribution ($\mu = 0, \sigma = 1$) and the TS distribution could be explained by statistical fluctuations. To determine whether the null hypothesis can be rejected, we calculated the p-value (p_v) for both samples, which rejects the null hypothesis if $p_v < 10^{-3}$ (W. H. Press et al. 2007). The final results indicate evidence of TeV emission from the whole sample, with a p-value of $p_v = 4 \times 10^{-92}$. After excluding the two most significant sources (Mrk 421 and Mrk 501), the obtained p-value was $p_v = 5 \times 10^{-17}$. These results reject the null hypothesis for both subsamples and confirm the detection of TeV emission from this population by HAWC.

Eighteen sources reported a test statistic $TS > 9$, which implies a significance $s > 3$. Five of these 18 sources had a significance $s > 5$ (TS > 25). Four of these five sources are BL Lac objects (Mrk 421, Mrk 501, PG 1218+304, and 1ES 1215+303), and one is a radio galaxy (M87). Moreover, 13 sources reported a marginal detection with $9 < TS < 25$, all of which are BL Lac objects. Additionally, seven of these sources were not previously reported as TeV emitters. These results represent a significant improvement compared to the previous survey, in which only five sources presented solid or marginal detections.

Due to HAWC's angular resolution ($\sim 2°$ for the lowest energy bins, which include most of the AGN data), there is a chance of cross-contamination between closely located sources. Among those sources with marginal or solid detections, some may have their emission contaminated by other objects. The blazars 1ES 1215+303 and PG 1218+304 are located only 0.8° from each other. Since both are known VHE emitters, a more detailed analysis is in preparation. The BL Lac object VER J0521+211 is located only 3.1° from the Crab Nebula, one of the brightest TeV sources in the sky. Moreover, the redshift of VER J0521+211 is under debate, so a more comprehensive analysis of this source will be carried out in future works. The BL Lac object RX J0648.7+1516 is situated in a crowded region of the TeV sky, surrounded by various sources, including Geminga. A more detailed analysis of this object will also be performed. Another interesting area for a complete multi-source analysis is the Coma cluster region, which includes several TeV-emitting BL Lacs such as W Comae, MS 1221.8+2452, PG 1218+304, and 1ES 1215+303.

5σ

5th check

It is worth mentioning that a spectral index of $\alpha = 2.5$ could differ significantly from the "real" index in some sources. However, since Wilks' theorem is only valid for one free parameter, fitting the spectral index would complicate the statistical analysis of the sample.

This work demonstrates HAWC's capacity to study VHE emission from AGN, particularly in characterizing their average TeV spectra. The increase in time exposure and the improvement in the HAWC analysis process (Pass 5) have produced excellent results, allowing for a better understanding of the characteristics of this population. Various studies can derive from this research, and since HAWC data acquisition is still ongoing, the availability of more data can substantially enhance any future analysis.

REFERENCES

Abe, K., Abe, S., Abhir, J., et al. 2025, Astronomy & Astrophysics, 694, A195

Abeysekara, A., Albert, A., Alfaro, R., et al. 2017a, The Astrophysical Journal, 843, 39

Abeysekara, A., Albert, A., Alfaro, R., et al. 2017b, The Astrophysical Journal, 841, 100

Abeysekara, A., Archambault, S., Archer, A., et al. 2017c, The Astrophysical Journal, 836, 205

Abeysekara, A., Albert, A., Alfaro, R., et al. 2019, The Astrophysical Journal, 881, 134

Abramowski, A., Acero, F., Aharonian, F., et al. 2012, The Astrophysical Journal, 746, 151

Acciari, V., Aliu, E., Beilicke, M., et al. 2008, The Astrophysical Journal, 684, L73

Acciari, V., Aliu, E., Beilicke, M., et al. 2010, The Astrophysical Journal Letters, 709, L163

Acciari, V., Arlen, T., Aune, T., et al. 2011, The Astrophysical Journal, 729, 2

Adams, C., Batista, P., Benbow, W., et al. 2022, The Astrophysical Journal, 932, 129

Aharonian, F., Akhperjanian, A., Beilicke, M., et al. 2003, Astronomy & Astrophysics, 403, L1

Ahn, C. P., Alexandroff, R., Prieto, C. A., et al. 2012, The Astrophysical Journal Supplement Series, 203, 21

Ahnen, M. L., Ansoldi, S., Antonelli, L., et al. 2016, Astronomy & astrophysics, 593, A91

Ahnen, M. L., Ansoldi, S., Antonelli, L. A., et al. 2017, Monthly Notices of the Royal Astronomical Society, 468, 1534

Ajello, M., Atwood, W., Baldini, L., et al. 2017, The Astrophysical Journal Supplement Series, 232, 18

Albert, A., Alfaro, R., Alvarez, C., et al. 2020, The Astrophysical Journal, 905, 76

Albert, A., Alvarez, C., Camacho, J. A., et al. 2021, The Astrophysical Journal, 907, 67

Albert, A., Alfaro, R., Alvarez, C., et al. 2022, The Astrophysical Journal, 929, 125

Albert, A., Alfaro, R., Alvarez, C., et al. 2024, arXiv preprint arXiv:2405.06050

Albert, J., Aliu, E., Anderhub, H., et al. 2006, The Astrophysical Journal, 642, L119

Albert, J., Aliu, E., Anderhub, H., et al. 2007, The Astrophysical Journal, 663, 125

Aleksić, J., Alvarez, E., Antonelli, L., et al. 2012, Astronomy & Astrophysics, 544, A142

Alfaro, R., Alvarez, C., Arteaga-Velázquez, J., et al. 2022, The Astrophysical Journal, 934, 158

Aliu, E., Aune, T., Beilicke, M., et al. 2011, The Astrophysical Journal, 742, 127

Archambault, S., Arlen, T., Aune, T., et al. 2013, The Astrophysical Journal, 776, 69

Bade, N., Beckmann, V., Douglas, N. G., et al. 1998, Astronomy & Astrophysics, 334, 459, doi: 10.48550/arXiv.astro-ph/9803204

Benkhali, F. A., Chakraborty, N., & Rieger, F. M. 2019, Astronomy & Astrophysics, 623, A2

Blandford, R., Meier, D., & Readhead, A. 2019, Annual Review of Astronomy and Astrophysics, 57, 467

Brien, S. O. 2019, in International Cosmic Ray Conference, Vol. 36, 36th International Cosmic Ray Conference (ICRC2019), 755, doi: 10.22323/1.358.0755

Cao, Z., Aharonian, F., An, Q., et al. 2024, The Astrophysical Journal Supplement Series, 271, 25

Collaboration, T. E. H. T., Akiyama, K., Alberdi, A., et al. 2019, The Astrophysical Journal Letters, 875, L1, doi: 10.3847/2041-8213/ab0ec7

Dominguez, A., Primack, J. R., Rosario, D., et al. 2011, Monthly Notices of the Royal Astronomical Society, 410, 2556

Feldman, G. J., & Cousins, R. D. 1998, Physical review D, 57, 3873

Finke, J. D., Razzaque, S., & Dermer, C. D. 2010, The Astrophysical Journal, 712, 238

Franceschini, A., & Rodighiero, G. 2017, Astronomy & Astrophysics, 603, A34

Franceschini, A., Rodighiero, G., & Vaccari, M. 2008, Astronomy & Astrophysics, 487, 837

Furniss, A., Worseck, G., Fumagalli, M., et al. 2019, The Astronomical Journal, 157, 41

Gorshkov, A., & Konnikova, V. 1993, Pis ma v Astronomičeskij žurnal, 19, 136

Heidt, J., Nilsson, K., Fried, J. W., Takalo, L. O., & Sillanpää, A. 1999, Astronomy and Astrophysics, 348, 113, doi: 10.48550/arXiv.astro-ph/9906129

Hinton, J. 2021, arXiv preprint arXiv:2111.13158

Mirzoyan, R. 2014, The Astronomer's Telegram, 5768, 1

Morris, S. L., Stocke, J. T., Gioia, I. M., et al. 1991, Astrophysical Journal, Part 1 (ISSN 0004-637X), vol. 380, Oct. 10, 1991, p. 49-65. Research supported by Smithsonian Scholarly Studies Program., 380, 49

Netzer, H. 2013, The physics and evolution of active galactic nuclei (Cambridge university press)

Ong, R. A., Paneque, D., Telescope, F. L. A., Collaboration, V., et al. 2010, The Astronomer's Telegram, 2486, 1

Padovani, P., Alexander, D., Assef, R., et al. 2017, The Astronomy and Astrophysics Review, 25, 2

Paiano, S., Landoni, M., Falomo, R., et al. 2017, The Astrophysical Journal, 837, 144

Pandey, A., & Stalin, C. 2022, Astronomy & Astrophysics, 668, A152

Press, W. H., Teukolsky, S. A., & Flannery, B. P. 2007, Numerical recipes: the art of scientific computing (Cambridge University Press)

Rector, T. A., Stocke, J. T., Perlman, E. S., Morris, S. L., & Gioia, I. M. 2000, The Astronomical Journal, 120, 1626

Sahakyan, N. 2020, Monthly Notices of the Royal Astronomical Society, 496, 5518

Saldana-Lopez, A., Domínguez, A., Pérez-González, P. G., et al. 2021, Monthly Notices of the Royal Astronomical Society, 507, 5144

Shaw, M. S., Romani, R. W., Cotter, G., et al. 2013, The Astrophysical Journal, 764, 135

Singh, K., Bisschoff, B., van Soelen, B., et al. 2019, Monthly Notices of the Royal Astronomical Society, 489, 5076

Valverde, J., Horan, D., Bernard, D., et al. 2020, The Astrophysical Journal, 891, 170

Weistrop, D., Shaffer, D., Hintzen, P., & Romanishin, W. 1985, Astrophysical Journal, Part 1 (ISSN 0004-637X), vol. 292, May 15, 1985, p. 614-619., 292, 614